

Supporting Information Methods S1

Article title: Big genera in Leguminosae: global distribution and taxonomic changes through time

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Contents

Reproducing the results

• Usage	1
• Extracting all Leguminosae/Fabaceae species	2
• Searching for the big legume genera and their nomenclatural data	3
• Plotting global distribution maps and graphics of the big legume genera	4

Usage

We have created **expowo** to automatize the process of extracting data from Kew's, Plants of The World Online database using R. Our goal is to pull species lists with all associated data of any vascular plant family. The results present in the paper makes uses of this new R package. If you do not already have it, install **expowo** from GitHub and load it with:

```
install.packages("devtools")
devtools::install_github("DBOSlab/expowo")
library(expowo)
```

Here we will show how to extract the data used in this paper, and then use this data to plot global distribution maps of species richness and the species accumulation curves.

Extracting all Leguminosae/Fabaceae species

To create the dataset with all legume species available at Plants of the World Online, use the function `powoSpecies`:

```
expowo::powoSpecies(family = "Fabaceae",
                    hybrid = FALSE,
                    verbose = TRUE,
                    save = TRUE,
                    dir = "results_powoSpecies/",
                    filename = "Fabaceae_search")
```

Searching for the big legume genera and their nomenclatural data

To search for the big genera, use the function `megaGen` and set the threshold to 500 species and the argument `family` to "Fabaceae".

```
mega_Fab <- expowo::megaGen(family = "Fabaceae",
                           thld = 500,
                           verbose = TRUE,
                           save = TRUE,
                           dir = "results_megaGen/",
                           filename = "legume_big_genera_species")
```

With this dataset, it was possible to filter and gather nomenclatural information. The custom R script used to do so is present here:

```
# Load required packages
library(magrittr)
library(dplyr)
library(ggplot2)
library(cowplot)
library(viridis)

# Import the file
list.files()
df <- read.csv("legume_big_genera_species.csv")

df <- apply(df, 2, function(x) gsub("^$", NA, x))
df <- data.frame(df)

# Creating new columns for the year and corrected year of publication
df <- df %>% tibble::add_column(year = NA, .after = "publication")
df <- df %>% tibble::add_column(year_corrected = NA, .after = "year")
df <- df %>% tibble::add_column(number_synonyms = NA, .after = "accepted_name")

# Extracting the year from the full publication information
# Clean examples such as "non C. caerulea Jacq.", "non Engelh. (1898)"
df$publication <- gsub(",\\snon\\s[[:upper:]].*", "", df$publication)

# Get examples of not validly published names
tf <- grepl("[0-9][ ]", df$publication)
```

```

p <- unique(gsub(".*[)]", "\\s", "", df$publication[tf]))

tf_inval <- !grepl(paste0(p, collapse = "|"), df$publication)

df$year[tf_inval] <- gsub("\\scontrary.*", "", df$publication[tf_inval])
df$year[tf_inval] <- gsub(".*[()|[]].*|.*\\.\\s|.*/", "", df$year[tf_inval])
df$year[tf_inval] <- gsub("unknown\\spublication|publication", "unknown",
  df$year[tf_inval])
df$year[!tf_inval] <- gsub(".*[()|.*/", "", df$publication[!tf_inval])
df$year[!tf_inval] <- gsub(".*[.]\\s|[]].*/", "", df$year[!tf_inval])
df$year[!tf_inval] <- gsub("unknown\\spublication|publication", "unknown",
  df$year[!tf_inval])

# Extracting the correct year for species with synonyms
# Fill corrected year for species without synonyms
tf_accepted <- df$status %in% "Accepted"
# Get the accepted names for species without any synonym
tftf <- !df$scientific_name[tf_accepted] %in% df$accepted_name
df$year_corrected[tf_accepted][tftf] <- df$year[tf_accepted][tftf]

# Count number of synonyms
df$number_synonyms[tf_accepted][tftf] <- 0
df$number_synonyms[tf_accepted][!tftf] <- table(df$accepted_name)

# Extracting the correct year for species with synonyms
n_au <- unique(df$accepted_name[!is.na(df$accepted_name)])
for (i in seq_along(n_au)) {

  tf <- df$scientific_name[tf_accepted] %in% n_au[i]
  year_acc <- df$year[tf_accepted][tf]

  tftf <- df$accepted_name[tf_inval] %in% n_au[i]
  temp <- df[tf_inval, ][tftf, ]
  temp <- temp[!grepl("\\svar[.]|\\sf[.]\\s|\\ssubsp[.]\\s",
    temp$scientific_name), ]
  if(length(temp$year) == 0) {
    year_syn <- NA
  } else {
    year_syn <- min(temp$year)
  }

  if (year_acc < year_syn |
    is.na(year_syn)) {
    df$year_corrected[tf_accepted][tf] <- year_acc
  } else {
    df$year_corrected[tf_accepted][tf] <- year_syn
  }
}

# Identify each change in the corrected year of species discovery
df$year <- as.numeric(df$year)
tf <- is.na(df$year_corrected)

```

```

df$year_corrected[!tf] <- as.numeric(df$year_corrected[!tf])

df$year <- as.numeric(df$year)
df$year_corrected <- as.numeric(df$year_corrected)

tf <- df$year == df$year_corrected

df <- df %>% tibble::add_column(year_changed = NA, .after = "year_corrected")
df$year_changed[tf] <- "no"
df$year_changed[!tf] <- "yes"

# Final spreadsheet including the input df with only the accepted species and
# associated corrected year
df_accepted <- df %>%
  filter(status == "Accepted") %>%
  select(-c("status", "accepted_name"))

write.csv(df_accepted, "data_to_plot.csv")

```

Plotting global distribution maps and graphs of the big legume genera

To create maps for each big legume genera with “rocket” color scale from Viridis, which composed the Figures 1-4, run the following R code:

```

# To create one map for each big genera according to political countries
expowo::powoMap(inputdf = mega_Fab,
  botctrs = FALSE,
  distcol = "native_to_country",
  taxclas = "genus",
  verbose = FALSE,
  save = TRUE,
  vir_color = "rocket",
  bre_color = NULL,
  leg_title = "SR",
  dpi = 600,
  dir = "results_powoMap/",
  filename = "Fab_global_richness_country_map",
  format = "jpg")

# To create one map for each big genera according to botanical subdivision
expowo::powoMap(inputdf = mega_Fab,
  botctrs = TRUE,
  distcol = "native_to_botanical_countries",
  taxclas = "genus",
  verbose = FALSE,
  save = TRUE,
  vir_color = "rocket",
  bre_color = NULL,

```

```

leg_title = "SR",
dpi = 600,
dir = "results_powoMap/",
filename = "Fab_global_richness_botcountry_map",
format = "jpg")

```

After running the previous steps of this section, use the dataframe “df_accepted” to plot the accumulation of species richness for each big legume genera.

Plot figures

```

scales::show_col(viridis(10, option = "magma"))
cols <- c("#180F3EFF", "#F1605DFF")

tax_group <- unique(df_accepted$genus)
for (i in seq_along(tax_group)) {

  temp_df_accepted <- df_accepted[df_accepted$genus %in% tax_group[i], ]
  l <- length(temp_df_accepted$species)

  plot <- ggplot(temp_df_accepted, aes(year, colour = year_changed))+
    #geom_vline(xintercept = 1850, colour = "red", linetype="dashed", size = 1,
    alpha=0.6)+
    stat_ecdf(linewidth = 1.5, alpha=0.6, geom = "step") +
    theme_bw() +
    ylab(eval(bquote(expression(
      bold("Accumulation of species discovery in")~bolditalic(. (tax_group[i])))))) +
    theme(axis.title.x=element_blank()) +
    scale_color_manual(values = cols) +
    scale_y_continuous(breaks=scales::pretty_breaks(n=5),
      labels=scales::percent_format(accuracy = 1)) +
    scale_x_continuous(breaks=c(1753, 1800, 1850, 1900, 1950, 2000,
      max(df$year[!is.na(df$year)]))) +
    theme(axis.title.y=element_text(size=14, margin=margin(0,12,0,0))) +
    theme(axis.text.x=element_text(size=14)) +
    theme(axis.text.y=element_text(size=14)) +
    theme(legend.title = element_text(size=14)) +
    theme(legend.position = c(0.2, 0.8),
      legend.key = element_rect(linewidth=10, linetype='blank'),
      legend.text = element_text(size=14),
      legend.key.size = unit(1, "cm"),
      legend.title = element_text(size=14)) +
    guides(alpha='none', colour=guide_legend("Publication year corrected"),
      size=14) +
    annotate("text", x = 2020, y = 1.02,
      label = paste(l, " spp."),
      colour = "black", alpha=0.6)

  save_plot(paste0("cumulative_discovery_", tax_group[i], ".jpg"), plot,
    ncol = 1, nrow = 1,
    base_height = 8.5,

```

```
    base_aspect_ratio = 1.3,  
    base_width = NULL)  
}
```